

SUPPLY CHAIN INTELLIGENCE

Inventory Health · Sales Performance · Demand Prediction

PROJECT PRESENTATION BRIEF

A descriptive source document for turning the project into a recruiter-facing presentation.

PYTHON / MYSQL / POWER BI / SCIKIT-LEARN

How to use this brief

Use each section as a presentation topic. The document explains what was done, why it matters, what to say aloud, and which limitations should be acknowledged. It is a portfolio project—not a deployed inventory or forecasting system.

Scope note: all results are specific to the supplied dataset and the current workflow. Reconcile the final notebook, CSV, SQL outputs and dashboard before posting.

01 | Business Problem & Dataset Overview

Business problem

Supply-chain teams balance product availability against the cost and operational risk of holding inventory. Low inventory can create replenishment pressure; excessive stock can tie up working capital. Sales also vary by product, region, warehouse, supplier, price and promotion status.

This project asks: **How can historical sales, inventory and replenishment signals be analyzed together to improve operational visibility?** As an extension, it explores whether a regression model can estimate Units_Sold from available features.

Dataset snapshot

91,250	16	50	5	4
records	columns	unique SKUs	warehouses	regions

The final dataset covers 2024-01-01 to 2024-12-30. Core fields include Date, SKU_ID, Warehouse_ID, Supplier_ID, Region, Units_Sold, Inventory_Level, Supplier_Lead_Time_Days, Reorder_Point, Order_Quantity, Unit_Cost, Unit_Price, Promotion_Flag and Demand_Forecast. Revenue and Inventory_Value are derived measures.

Business questions

- Which regions and SKUs contribute the most units sold or revenue?
- Where do inventory records sit at or below reorder thresholds?
- Which suppliers have longer average lead times?
- How do sales records differ between promotion and non-promotion periods?
- Can a model estimate units sold on later-dated records?

Key message: connect technical analysis to practical questions about sales performance, inventory health and replenishment signals.

02 | Data Preparation & Derived Measures

The Pandas workflow inspects the dataset, reviews structure and quality, prepares date columns, and creates business-friendly measures. The final CSV was checked as 91,250 rows and 16 columns. SKU labels such as SKU_18 are identifiers, not descriptions of actual products.

Step	What was done	Why it matters
Data-quality review	Check dimensions, column names, data types, missing values, duplicates and date coverage.	Confirms the data is suitable for analysis.
Date handling	Convert Date to datetime; derive Month and DayOfWeek for ML.	Supports time trends and calendar features.
Revenue	$\text{Revenue} = \text{Units_Sold} \times \text{Unit_Price}$.	Creates a consistent row-level sales value.
Inventory value	$\text{Inventory_Value} = \text{Inventory_Level} \times \text{Unit_Cost}$.	Estimates the cost value represented by each snapshot row.
Lag/rolling features	Include Sales_Lag_1, Sales_Lag_7 and Sales_Rolling_7 when present.	Adds recent sales context; must be computed without future information.

Important inventory definition

Inventory is recorded repeatedly over time. Summing Inventory_Level across every day does not mean current inventory. Latest-inventory KPIs should use only the latest recorded date (2024-12-30). The latest snapshot previously checked contained **114,019 units** and **1,404,610.84** in inventory value; reconcile these figures against the final CSV and dashboard before publishing.

Likewise, $\text{Inventory_Level} \leq \text{Reorder_Point}$ counts records at or below the threshold. It is a replenishment signal, not proof that a stockout occurred. Stockout_Flag is intentionally not included in the final dataset.

03 | Python EDA in Jupyter Notebook

Exploratory Data Analysis (EDA) establishes a baseline understanding of the data before SQL reporting and machine learning. The notebook summarizes the dataset, compares groups, and uses charts to make patterns easier to see.

EDA area	What was explored	How to explain it
Dataset profile	Rows, columns, data types, dates and quality checks.	Establishes the coverage and reliability of the analysis.
Sales patterns	Units sold by date, month, region and SKU.	Shows where sales volume is concentrated and how it changes.
Revenue	Revenue from units sold multiplied by unit price.	Adds a value-based view alongside volume.
Inventory health	Inventory levels, reorder points and inventory value.	Highlights records that may require replenishment review.
Promotion comparison	Average units sold for promotion-flagged vs non-promotion records.	Descriptive association only; not proof of causality.
Visual checks	Line plots, bar charts and grouped comparisons.	Helps identify trends, differences and unusual patterns.

Selected descriptive results

- Total Units Sold: **1,829,979**.
- Total Revenue: **33,426,337.22**; currency is not specified in the source data.
- Records at or below reorder point: **5,041**; this is a record count, not a unique-SKU count.
- Average Units Sold was approximately **24.91** on promotion-flagged records versus **19.50** on non-promotion records.

Key message: EDA gives context to the later SQL and ML work, but group differences should not be described as causal effects.

04 | MySQL: All 14 Analysis Areas

The 14 SQL queries translate business questions into repeatable analysis. They cover aggregation and sorting as well as CASE logic, subqueries, CTEs, conditional aggregation and window functions.

#	Business question / analysis	SQL concept practiced
1	Which region has the highest sales volume?	SUM, GROUP BY, ORDER BY
2	Which warehouse has the highest summed inventory across records?	SUM, GROUP BY; snapshot caveat
3	Which SKUs have the most records at/below reorder point?	WHERE, COUNT, GROUP BY
4	Which suppliers have the longest average lead times?	AVG, grouping, sorting
5	Which SKUs generate the highest revenue?	Calculated expression, aggregation, LIMIT
6	Which individual records have the highest inventory?	Sorting and top-N
7	How can records be classified High Risk or Normal?	CASE expression
8	Which SKUs exceed average total sales per SKU?	Subquery, HAVING
9	Which warehouses exceed average summed inventory?	Nested aggregation, HAVING
10	Which SKUs have highest inventory value?	CTE, calculated value
11	Which warehouses have the highest share of low-inventory records?	CTE, CASE, percentage
12	What is each SKU's revenue rank within its region?	RANK() OVER, PARTITION BY
13	What share of regional revenue comes from each SKU?	Window aggregation, percentage
14	How does average sales differ by promotion status and region?	Conditional aggregation

Interpretation guardrail

Warehouse queries that sum Inventory_Level across the whole year describe accumulated snapshot values—not current on-hand stock. For a current-inventory KPI, filter to the latest date. Reorder-point queries flag records at/below a threshold; they do not confirm stockouts.

05 | Power BI Dashboard Plan & Key Metrics

The dashboard concept is a one-page **Supply Chain Intelligence** view. It should show headline KPIs first, then sales trends, regional revenue, SKU performance and replenishment signals. A restrained navy/teal/white palette and a small number of slicers keep it readable.

Dashboard element	Definition / intended use
Total Units Sold	SUM of Units_Sold across the selected scope.
Total Revenue	SUM of Revenue, derived from Units_Sold × Unit_Price.
Latest Inventory Units	SUM Inventory_Level on the latest recorded date only.
Latest Inventory Value	SUM Inventory_Value on the latest recorded date only.
Low-Inventory Records	Count of rows where Inventory_Level ≤ Reorder_Point; not stockouts.
Monthly sales trend	Units_Sold by month.
Revenue by region	Compare revenue across East, North, South and West.
Latest inventory by warehouse	Compare latest-date inventory for five warehouses.
Top 10 SKUs by revenue	Rank by revenue; SKU_ID remains an identifier.
Low-inventory records by warehouse	Compare counts or percentages at/below reorder point.
Slicers	Region and Warehouse; date filters must not distort latest-snapshot KPIs.

Dashboard quality checks

- Reconcile KPI totals with Python and SQL before publishing.
- Clearly distinguish full-period totals from latest-date snapshot values.
- Use readable labels and avoid misleading truncated axes.
- Test slicers and verify that the latest-inventory measures behave as intended.

06 | Random Forest Training Workflow & Evaluation

The regression task estimates Units_Sold from product, warehouse, supplier, region, inventory, pricing, promotion, demand forecast, lag/rolling and calendar features. This is an exploratory forecasting baseline—not an automated replenishment decision system.

Stage	Implementation
Target	Units_Sold
Feature setup	Use available columns from the feature list; derive Month and DayOfWeek from Date.
Chronological split	Train on dates before 2024-10-01; test on dates from 2024-10-01 onward.
Preprocessing	OneHotEncoder(handle_unknown='ignore') for categorical fields; pass numeric fields through.
Model	RandomForestRegressor with 100 trees, random_state=42 and n_jobs=-1.
Evaluation	Predict on the holdout set and calculate MAE, RMSE and R².
Review	Compare actual vs predicted sales and inspect permutation importance.

Notebook evaluation output

2.26	2.84	0.7857	22,750
MAE · units	RMSE · units	R²	test records

MAE 2.26 means absolute prediction error averaged about 2.26 units. **RMSE 2.84** penalizes larger errors more strongly. **R² 0.7857** summarizes fit relative to a mean-based reference on this holdout set; it does not guarantee future performance.

Validation before posting

Confirm lag/rolling features were calculated in chronological order within the correct SKU/warehouse series and do not use future values. Verify that Demand_Forecast is available at prediction time and was not constructed using the target or future data. Compare Random Forest with a simple baseline such as last-period sales or a rolling mean.

07 | Feature Importance & Responsible AI Usage

The reported permutation-importance output places **Demand_Forecast** far above the other displayed features (about 4.07 in the reported scoring units), followed by **Sales_Rolling_7** (about 0.042) and **Promotion_Flag** (about 0.020). Smaller values were reported for Sales_Lag_1, Sales_Lag_7, DayOfWeek, Region and Unit_Cost; Month was near zero and Order_Quantity slightly negative.

Permutation importance measures how model performance changes when a feature is shuffled. It is not a causal explanation or a percentage contribution. A slightly negative score can arise from sampling or scoring variation; it should not be interpreted as a meaningful negative business effect.

AI-assisted workflow

AI-assisted activity	How it supported the work	Verification
Troubleshooting	Helped interpret warnings and diagnose execution errors such as an undefined df_ml variable.	Rerun corrected cells and inspect notebook outputs.
Workflow guidance	Helped organize data preparation, SQL, model evaluation and feature-importance steps.	Use the final notebook and SQL script as source of truth.
Concept explanation	Helped explain MAE, RMSE, R ² , encoding and permutation importance.	Keep explanations consistent with implemented code.
Presentation support	Helped structure the recruiter-facing narrative and caveats.	Do not claim deployment or unmeasured business impact.

Suggested disclosure

"I used generative AI as a coding and learning assistant to troubleshoot errors, clarify machine-learning concepts and structure the project narrative. I validated outputs in Python/Jupyter and MySQL. AI assistance did not replace verification or business judgment."

The Random Forest model was fitted by the scikit-learn code in the notebook; do not say AI trained the model on its own.

08 | Limitations & Next Steps

Limitations to state clearly

- The dataset covers one year; results may not generalize to another business or season.
- Inventory values repeat across daily snapshots; full-year sums are not current stock.
- At/below-reorder-point records are a risk indicator, not confirmed stockouts; Stockout_Flag is intentionally absent.
- Promotion comparisons are descriptive and do not establish causality.
- The reported model has not yet been benchmarked against a simple forecasting baseline.
- Demand_Forecast and lag/rolling features require leakage checks before real-world use.
- The model has not been deployed or tested in a live replenishment process.

Practical next steps

Priority	Action	Why
1	Rerun the notebook top-to-bottom and save the final version.	Reproducibility and clean outputs.
2	Audit lag/rolling generation and forecast availability.	Prevent future-information leakage.
3	Compare Random Forest with last-period and rolling-mean baselines.	Check whether ML adds value.
4	Report errors by SKU, warehouse, region and month.	Find weak segments and patterns.
5	Reconcile Python, SQL and Power BI totals.	Build trust in reported KPIs.
6	If operationalized, monitor errors and retain human review.	Avoid blind automation.

Final project summary

This project demonstrates an end-to-end analytics workflow: data preparation and EDA in Python, 14 business-focused MySQL analyses, a Power BI dashboard plan, and a Random Forest regression pipeline evaluated on later-dated records. The portfolio value is the connection between technical work, business questions, clear communication and responsible limitations.

Appendix | Suggested Presentation Flow

Use these eight topics as your slide sequence. Each row is the core point the audience should remember.

Slide	Focus	Core message
1	Business problem + dataset	I brought sales, inventory and replenishment signals together to study operational visibility and explore unit-sales prediction.
2	Preparation + derived measures	I checked the data and created Revenue and Inventory_Value for sales and inventory-value analysis.
3	Python EDA	EDA helped me understand trends, group differences, inventory thresholds and promotion-related patterns.
4	14 SQL areas	The SQL work progressed from aggregation to CASE logic, subqueries, CTEs, window functions and conditional aggregation.
5	Power BI dashboard	The one-page view brings headline KPIs and sales/replenishment visuals together while distinguishing latest snapshots from full-period totals.
6	Random Forest + evaluation	I used a chronological holdout, categorical preprocessing and a Random Forest regressor, then evaluated prediction errors.
7	Feature importance + AI	Demand_Forecast dominated permutation importance in this run; AI supported troubleshooting and explanation, while outputs were verified.
8	Limitations + next steps	Before real use, I would audit leakage, compare baselines, inspect segment-level errors and reconcile metrics.

Suggested LinkedIn caption

Built a Supply Chain Inventory & Sales Performance Analysis project using Python, MySQL, Power BI and scikit-learn. I explored 91,250 records, created derived sales and inventory-value measures, completed 14 SQL analyses, and trained a Random Forest regression model to estimate units sold. On a chronological holdout of 22,750 records, the model reported MAE 2.26, RMSE 2.84 and R^2 0.7857. I also reviewed permutation feature importance and documented validation steps, including leakage checks and baseline comparisons. I used generative AI as a coding and learning assistant for troubleshooting and explanation, while verifying results in the notebook. This is a portfolio project, not a production forecasting deployment.

Before posting, verify the final notebook is rerun and saved, and reconcile latest-inventory values against the final CSV and dashboard.